Loss Functions with Explicit Node Indices

Here, v indexes a node and i indexes a class. Every node-specific quantity carries the node index explicitly. Thus, the target, prediction, teacher output, and input appearing in a summand all belong to the same node v.

Direct MLP training. For a labeled node v, let $\mathbf{x}_v$ be its attributes, $\mathbf{t}_v$ its soft target, and $\mathbf{p}_v$ the MLP prediction:

$$\mathbf{p}_v = \mathrm{softmax}\big(f(\mathbf{x}_v)\big), \qquad v \in \mathcal{V}_L. \tag{1}$$

The cross-entropy objective is

$$\mathcal{L} = -\frac{1}{|\mathcal{V}_L|} \sum_{v \in \mathcal{V}_L} \sum_{i=1}^{C} t_{v,i} \log p_{v,i}, \tag{2}$$

where $t_{v,i}$ and $p_{v,i}$ are the class-i components of $\mathbf{t}_v$ and $\mathbf{p}_v$.

Graph-teacher distillation. Let $\mathbf{o}_v$ denote the GCN teacher's logit vector for node v. The teacher is trained on labeled nodes using the same soft targets:

$$\mathcal{L}_1 = -\frac{1}{|\mathcal{V}_L|} \sum_{v \in \mathcal{V}_L} \sum_{i=1}^{C} t_{v,i} \log \left[\mathrm{softmax}(\mathbf{o}_v)\right]_i . \tag{3}$$

After the teacher is frozen, it produces a distillation distribution for every node:

$$\boldsymbol{\pi}_v = \mathrm{softmax}\left(\frac{\mathbf{o}_v}{\tau}\right), \qquad v \in \mathcal{V}. \tag{4}$$

The MLP student is trained on all nodes by minimizing

$$\mathcal{L}_2 = -\frac{\tau^2}{N} \sum_{v \in \mathcal{V}} \sum_{i=1}^{C} \pi_{v,i} \log \left[\mathrm{softmax}\left(\frac{f(\mathbf{x}_v)}{\tau}\right)\right]_i . \tag{5}$$

Here, $\pi_{v,i}$ is the class-i component of the teacher distribution for node v. The explicit subscripts make the node-to-term correspondence visible, at the cost of longer formulas.